

End Sem Paper and Solutions

Instructions

1. All questions carry equal marks.
2. Read all questions carefully before putting pen to paper!
3. You have 3 hours to complete this examination.
4. One page (or at most two pages) should be enough to answer each question!
5. Complete the solution to each question on contiguous pages.

Question 1

- a. Show that $125x^3 + 9y^3 = 15z^3$ has no solutions with x, y, z rational numbers except $(x, y, z) = (0, 0, 0)$.
- b. Does this equation have non-trivial solutions in p -adic integers for large enough p ? Justify your answer.

Solution. If x and y are non-zero then $v_3(x)$ and $v_3(y)$ are integers. By comparing divisibility of these integers by 3 we see that

$$v_3(125x^3) = 3v_3(x) \neq 2 + 3v_3(y) = v_3(9y^3)$$

It follows that $v_3(27x^3 + 49y^3) = \min\{3v_3(x), 2 + 3v_3(y)\}$. The given equation then implies that

$$\min\{3v_3(x), 2 + 3v_3(y)\} = v_3(125x^3 + 9y^3) = v_3(15z^3) = 1 + 3v_3(z) \text{ — equation (1)}$$

It follows that $z \neq 0$ and $v_3(z)$ is an integer. On the other hand, divisibility by 3 again gives us the inequalities:

$$\begin{aligned} v_3(15z^3) &= 1 + 3v_3(z) \neq 3v_3(x) \\ v_3(15z^3) &= 1 + 3v_3(z) \neq 2 + 3v_3(y) \end{aligned}$$

This shows that the equation (1) cannot hold in this case. For the same reason, we cannot have one of x or y as 0 and the other non-zero.

Thus, the only solution in $\mathbb{Q}_3$ is $(0, 0, 0)$.

Note that this is an equation of “Euler-Fermat” type in 3 variables. Hence, it has non-trivial solutions in $\mathbb{F}_p$ for all p sufficiently large.

Suppose that (x_0, y_0, z_0) is such a solution in $\mathbb{F}_p$ for some prime $p > 125$. If $x_0 \neq 0$ then we can apply Hensel’s lemma to the equation $125x^3 + 9y_1^3 - 15z_1^3$ where y_1, z_1 are elements in $\mathbb{Z}$ that reduce to y_0, z_0 modulo p respectively. A similar argument can be used if $y_0 \neq 0$. Thus, we get a non-trivial solution in $\mathbb{Z}_p$ for all large enough p .

Question 2

- a. Show that 6 has a cube-root in $\mathbb{Z}_5$, the ring of 5-adic integers.
- b. Give an example of an integer $-12 \leq a \leq 12$ such that $a^3 - 6$ is divisible by 25.

Solution. Let us represent elements of $\mathbb{F}_5$ as $\{-2, -1, 0, 1, 2\}$.

We note that 1 is a root of $X^3 - 6$ in $\mathbb{F}_5$.

We put $x_0 = 1$ and use the iteration $x_{n+1} = x_n - f(x_n)/f'(x_0)$ where $f(X) = X^3 - 6$ and $f'(X) = 3X^2$. By Hensel's lemma the sequence $(x_n)_n$ converges in $\mathbb{Z}_5$ to a root of f .

We note that $f'(x_0) = -2$ when $x_0 = 1$, so $1/f'(x_n)$ can be replaced by 2, since $2 \cdot (-2) = 1$ in $\mathbb{F}_5$.

Thus, $x_1 = 1 - 2(1^3 - 6) = 1 + 10 = 11$ modulo 25. We easily check that

$$11^3 - 6 = 1331 - 6 = 1325$$

is divisible by 25.

Question 3

- a. Find the smallest n for which $\mathbb{F}_{2^n}$ contains a primitive 7-th root of unity.
- b. Give an example of a homogeneous polynomial of degree 6 in 2 variables over $\mathbb{F}_2$ that has no *non-trivial* solutions.

Solution. We check in $(\mathbb{Z}/\langle 7 \rangle)^*$ that the powers of 2 are

$$2^1 = 2, 2^2 = 4, 2^3 = 1$$

As seen earlier, this means that the smallest n as required is $n = 3$.

In particular, this means that the polynomial $X^6 + X^5 + X^4 + X^3 + X^2 + X + 1$ has no roots in $\mathbb{F}_2$.

It follows easily that the polynomial $X^6 + X^5Y + X^4Y^2 + X^3Y^3 + X^2Y^4 + XY^5 + Y^6$ has no non-trivial solutions with X, Y in $\mathbb{F}_5$.

Question 4

- a. Write a polynomial in 2 variables over $\mathbb{F}_5$ such that it takes the value 0 at all (X, Y) , *except* at $(0, 1)$ and $(1, 0)$, where it takes the values 1 and -1 , respectively.
- b. What is the least possible degree of such a polynomial?

Solution. The unique polynomial in $(\mathbb{Z}/\langle 5 \rangle)[x, y]/(x^5 - x, y^5 - y)$ which has this property is

$$g(x, y) = (1 - x^4)(1 - (y - 1)^4) - (1 - (x - 1)^4)(1 - y^4)$$

We note that the term of degree 8 in this polynomial is 0.

The polynomials in the above ring have x degree at most 4 and y degree at most 4. Thus, the polynomial of least degree in this ring with this property will have degree at most 8.

This proves that $g(x, y)$ which has degree 7 is the polynomial of lowest degree with this property.

Question 5

In this question we use ζ to denote a 6-th root of unity in some field K of characteristic 0 and $R = \mathbb{Z}[\zeta]$ is the subring of K generated by ζ .

- a. Show that $1 - \zeta^3$ is not a unit in R .
- b. Show that $1 - \zeta^2$ is not a unit in R .
- c. Show that $1 - \zeta$ is a unit in R .

Solution. Since R is a ring of algebraic integers, we note that a unit in R that lies in $\mathbb{Z}$ is ± 1 .

We note that ζ^3 is a primitive square root of 1 in K . Hence, $\zeta^3 = -1$. So $1 - \zeta^3 = 2$ which is not a unit in R .

We note that ζ^2 is a primitive cube root of 1 in K . Hence $1 + \zeta^2 + \zeta^4 = 0$. It follows that

$$(1 - \zeta^2)(1 - \zeta^4) = 1 - (\zeta^2 + \zeta^4) + 1 = 3$$

which is not a unit. It follows that $(1 - \zeta^2)$ is not a unit.

We note that $\zeta^5 = -\zeta^2$ and $\zeta^4 = -\zeta$. Thus $1 - \zeta + \zeta^2 = 0$. It follows that

$$(1 - \zeta)(1 - \zeta^5) = 1 - (\zeta - \zeta^2) + 1 = 1$$

It follows that $(1 - \zeta)$ is a unit.

Question 6

In each case below give an example of the required type or say that it does not exist.

- a. A Dirichlet character χ with conductor 2.
- b. A Dirichlet character χ with conductor 7 such that $\chi(2) = 1$.
- c. A Dirichlet character χ with conductor 5 such that $\chi(2) = 1$.
- d. A Dirichlet character χ with conductor 20 with values in $\{0, \pm 1\}$.

Solution.

- a. Since $(\mathbb{Z}/\langle 2 \rangle)^* = \{1\}$, every character with modulus 2 is the trivial character and its conductor is 1. So, there is no such character.
- b. We note that $(\mathbb{Z}/\langle 7 \rangle)^*$ is generated by 3 and $3^2 = 2$ in this group. It follows that we can define $\chi(3) = -1$ to get $\chi(2) = 1$.
- c. We note that $(\mathbb{Z}/\langle 5 \rangle)^*$ is generated by 2. It follows that if $\chi(2) = 1$, then $\chi(a) = 1$ for *all* a with $\gcd(a, 5) = 1$. In other words, χ has conductor 1. Hence, there is no such character.
- d. We can define the character using $20 = 4 \times 5$. From the Chinese Remainder theorem, we see that:
 - 11 is such that it is 3 modulo 4 and 1 modulo 5.
 - 13 is such that it is 1 modulo 4 and 3 modulo 5.

Thus, we define $\chi(11) = -1 = \chi(13)$. It follows that the conductor does not divide 10 and does not divide 4. Hence the conductor must be 20.